Chapter 2: Some non-parametric procedures — Solutions

STA310 Survival Analysis (Master's in Applied Biostatistics)

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Chapter 2 covers non-parametric (distribution-free) methods for summarising survival data: estimating the survivor, hazard and cumulative hazard functions, estimating the median and other percentiles, and comparing two or more groups with the log-rank, Wilcoxon and Peto–Peto tests. These exercises emphasise interpretation and application rather than the underlying derivations.

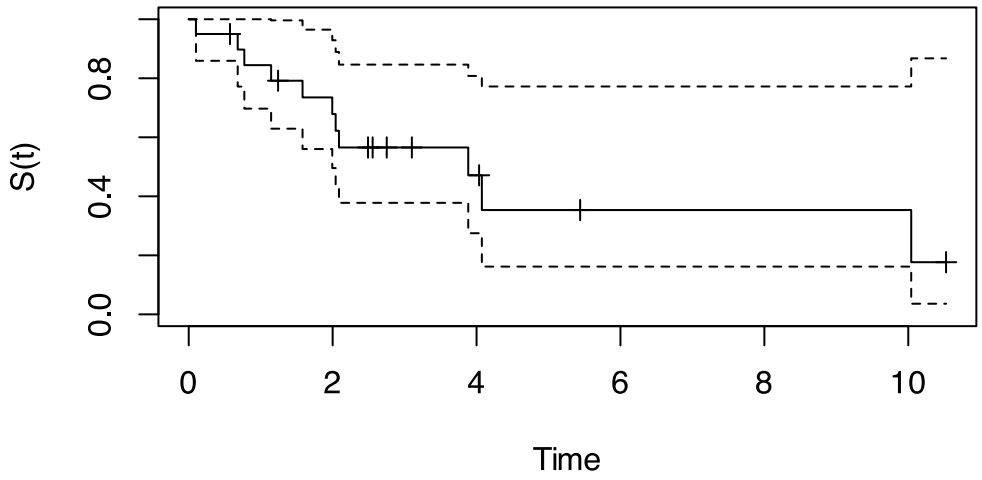
1 Estimating the survivor function (Book §2.1–2.2)

1.1 The Kaplan–Meier curve

Type: Interpret figure

Builds on: Chapter 1

The figure shows a Kaplan–Meier estimate of the survivor function for a single sample with some censored observations. The estimate is a step function, and a number of time points are marked with small tick marks.



(a) Why does the estimate drop only at observed event times, and not at the censored times?
(b) What information do the tick marks convey?
(c) The largest observation is censored. What does this imply about the estimate beyond that time?

Solution.

(a) The Kaplan–Meier estimate is a product of survival probabilities $(n_j - d_j)/n_j$ at each event time. Between events no one dies, so the estimate is constant; censored times contribute information (they reduce the risk set) but do not themselves cause a drop.
(b) The tick marks show where an individual was censored — left the risk set without an observed event.
(c) The estimate is undefined beyond the largest (censored) time: it never reaches 0, because no event was observed after that point.

1.2 Computing a Kaplan–Meier estimate

Type: Concept & notation

Six patients are followed; their times (months) to an event, with censored times marked with "+", are:

$$3{,}4+{,}5{,}5{,}7{,}10+$$

(a) List the distinct event times and the number at risk just before each one.
(b) Compute the Kaplan–Meier estimate of $S(t)$ at each event time.
(c) What is the value of the estimate beyond 10 months?

Solution.

(a) Event times 3, 5 and 7. At risk just before $t = 3$: 6; before $t = 5$: 4 (one death at 3, and the $4+$ patient censored); before $t = 7$: 2.
(b) $\hat{S}(3) = 5/6 \approx 0.833$; $\hat{S}(5) = 0.833 \times (2/4) = 0.417$; $\hat{S}(7) = 0.417 \times (1/2) = 0.208$.
(c) The largest observation (10) is censored, so $\hat{S}(t)$ is undefined beyond 10 and never reaches 0.

1.3 Life table, Kaplan–Meier and Nelson–Aalen

Type: Method selection

(a) When is the life-table (actuarial) estimate of the survivor function the appropriate choice, and when would you prefer the Kaplan–Meier estimate?
(b) How do the Kaplan–Meier and Nelson–Aalen estimates relate to each other — which is the larger, and when are they very similar?
(c) Why does the life-table estimate lose information when individual event times are actually known?

Solution.

(a) The life table is suited to grouped data, where only the numbers of deaths and censored observations in each interval are known (e.g. interval-censored data). When the individual event times are known, the Kaplan–Meier estimate is preferred.
(b) The Nelson–Aalen estimate is always at least as large as the Kaplan–Meier estimate (because $e^{-x} \geq 1 - x$); the two are very similar except at the longest survival times, where the risk set is small.
(c) Grouping the times discards the exact event times, so the estimate is coarser (less precise) than one based on the individual times.

1.4 The precision of the estimate

Type: Concept & notation

From a Kaplan–Meier analysis, $\hat{S}(12 \text{ months}) = 0.65$ with a 95% confidence interval $(0.52, 0.78)$.

(a) Interpret this confidence interval in words.
(b) Explain why the standard error of $\hat{S}(t)$ — and hence the width of the confidence interval — generally increases as t increases.

Solution.

(a) We are 95% confident that the true probability of surviving beyond 12 months lies between 0.52 and 0.78. (It is a pointwise interval, valid at this single time.)
(b) At later times the estimate is based on fewer individuals (the risk set shrinks as people die or are censored), so there is more uncertainty. Greenwood's formula sums terms $d_j/[n_j(n_j - d_j)]$ that grow as n_j falls, so the standard error increases with time.

2 Estimating the hazard and the median (Book §2.3–2.5)

2.1 Estimating the cumulative hazard

Type: Concept & notation

Builds on: Chapter 1

From Chapter 1, the cumulative hazard is $H(t) = -\log S(t)$.

(a) Given a Kaplan–Meier estimate $\hat{S}(t)$, how would you estimate the cumulative hazard $\hat{H}(t)$?
(b) The Nelson–Aalen estimate of $H(t)$ is the sum $\sum_j d_j/n_j$ over the event times up to t. Explain in words what this sum accumulates.
(c) If a plot of the estimated cumulative hazard is roughly a straight line, what does this suggest about the hazard function?

Solution.

(a) $H(t) = -\log S(t)$ is a relationship between two *population* quantities — the true cumulative hazard and the true survivor function. To estimate the cumulative hazard, the same relationship is applied to the *estimated* survivor function: $\hat{H}(t) = -\log \hat{S}(t)$. In other words, $H(t)$ and $S(t)$ are the unknown population values, whereas $\hat{H}(t)$ and $\hat{S}(t)$ are their sample-based estimates (here, $\hat{S}(t)$ is the Kaplan–Meier estimate).
(b) It adds up the estimated probability of death d_j/n_j at each event time — the expected number of events accumulated up to time t (the cumulative risk).
(c) The slope of the cumulative hazard is the hazard, so a straight line means the hazard is roughly constant over that period.

2.2 Estimating the median survival time

Type: Concept & notation

A Kaplan–Meier estimate gives the following values of the survivor function:

t	10	20	30	40
$\hat{S}(t)$	0.72	0.55	0.41	0.30

(a) What is the estimated median survival time, and how is it read off the estimate?
(b) In a different study, the estimate never falls below 0.5 during the whole period of follow-up. What does this mean for estimating the median, and what could you report instead?

Solution.

(a) The median is the smallest time at which $\hat{S}(t) < 0.5$: here $\hat{S}(30) = 0.41 < 0.5$ (and $\hat{S}(20) = 0.55 > 0.5$), so the estimated median is 30.
(b) The median cannot be estimated, because survival never drops to 0.5 within the observed range. Report other percentiles, or the estimated survival probabilities at fixed time points instead.

2.3 Computing a Nelson–Aalen estimate

Type: Concept & notation

Builds on: Chapter 1

Seven patients are followed until an event (in months), with a censored time marked "+":

$$3{,}5{,}5{,}8+{,}10{,}12{,}15$$

(a) List the distinct event times and the number at risk just before each one.
(b) Compute the Nelson–Aalen estimate of the cumulative hazard, $\hat{H}(t)$, at each event time.
(c) Compute the Kaplan–Meier estimate $\hat{S}(t)$ at the same event times, and compare it with $\exp\{-\hat{H}(t)\}$. Which is the larger?
(d) What is the estimated median survival time?

Solution.

(a) Event times 3, 5, 10, 12 and 15. At risk just before each: 7, 6, 3, 2 and 1 (the $8+$ patient leaves the risk set between the two events at 5 and the event at 10).

(b) The Nelson–Aalen estimate accumulates d_j/n_j: $\hat{H}(3) = 1/7 = 0.143$; $\hat{H}(5) = 0.143 + 2/6 = 0.476$; $\hat{H}(10) = 0.476 + 1/3 = 0.810$; $\hat{H}(12) = 0.810 + 1/2 = 1.310$; $\hat{H}(15) = 1.310 + 1/1 = 2.310$.

(c) Kaplan–Meier: $\hat{S}(3) = 6/7 = 0.857$; $\hat{S}(5) = 0.857 \times 4/6 = 0.571$; $\hat{S}(10) = 0.571 \times 2/3 = 0.381$; $\hat{S}(12) = 0.381 \times 1/2 = 0.190$; $\hat{S}(15) = 0$. The corresponding $\exp\{-\hat{H}(t)\}$ values are 0.867, 0.621, 0.445, 0.270 and 0.099 — each is at least as large as the Kaplan–Meier estimate (because $e^{-x} \geq 1 - x$), so the Nelson–Aalen-based survivor estimate never lies below the Kaplan–Meier estimate.

(d) The median is the smallest time with $\hat{S}(t) < 0.5$: $\hat{S}(5) = 0.571 > 0.5$ and $\hat{S}(10) = 0.381 < 0.5$, so the estimated median is 10 months.

3 Comparing two or more groups (Book §2.6–2.9)

3.1 Choosing a test for two groups

Type: Method selection

You want to compare the survival experience of two treatment groups. Which of the log-rank, Wilcoxon and Peto–Peto tests would you use in each situation, and why?

(a) The two estimated survivor functions do not cross, and proportional hazards seems reasonable.
(b) You are mainly interested in differences in survival early on, when most events occur.
(c) The pattern of censoring differs between the two groups.

Solution.

(a) The log-rank test — it is the most powerful under proportional hazards.
(b) The Wilcoxon test — it weights each difference by the risk-set size n_j, so it down-weights the tail and is more sensitive to early differences.
(c) The Peto–Peto test — it performs better than the Wilcoxon test when the censoring pattern differs between groups.

3.2 The log-rank test and proportional hazards

Type: Assumptions

Builds on: Chapter 1

(a) In your own words, what does *proportional hazards* mean?
(b) Why is "the two estimated survivor functions do not cross" only a necessary, not a sufficient, check for proportional hazards?
(c) Using $S(t) = \exp\{-H(t)\}$ from Chapter 1, explain why proportional hazards implies that the two survivor functions never cross.

Solution.

(a) The hazard in one group is a constant multiple of the hazard in the other group at every time t: $h_1(t) = \psi\, h_2(t)$.
(b) Sample-based estimates can cross by chance even when the true hazards are proportional, and non-crossing curves do not guarantee proportional hazards. It is only a rough, informal check.
(c) If $h_1 = \psi h_2$ then $H_1 = \psi H_2$, so $S_1(t) = \exp(-H_1) = \exp(-\psi H_2) = \{S_2(t)\}^{\psi}$, which is either always above or always below $S_2(t)$; hence the curves never cross.

3.3 Interpreting a log-rank test

Type: Interpret R output

The output below compares the survival times of women with breast cancer, grouped according to whether a section of the tumour was positively or negatively stained.

```
survdiff(Surv(time, status) ~ stain)

                  N Observed Expected (O-E)^2/E (O-E)^2/V
stain=positive   13        5     9.57      2.18      3.52
stain=negative   32       20    15.43      1.35      3.52

 Chisq= 3.5  on 1 degrees of freedom, p= 0.06
```

(a) State the null hypothesis being tested.
(b) For the positively stained group, interpret the observed (5) and expected (9.57) numbers of deaths.
(c) What are the value of the test statistic, its degrees of freedom and the p-value? What do you conclude?

Solution.

(a) No difference in the survival experience (the survivor functions) of the two staining groups.
(b) Five deaths were observed, whereas 9.57 would be expected if the two groups had the same survival experience (the expected count is built up from the risk sets). Fewer deaths than expected suggests better survival in the positive group.

(c) $\chi^2 = 3.5$ on 1 degree of freedom, $p = 0.06$. This is slight evidence against the null hypothesis (between 0.05 and 0.1), not significant at the conventional 5% level — so at most weak evidence of a difference in prognosis.

3.4 Computing the log-rank statistic

Type: Risk set reasoning

Two groups of patients are followed until an event (in months); there is no censoring:

Group A: 5,9,13,17 Group B: 2,4,7,11

(a) List the ordered event times, and for each give the numbers at risk in groups A and B.
(b) For group A, compute the observed number of deaths O_A and the expected number $E_A = n_A d/(n_A + n_B)$ at each event time, and hence the totals O_A and E_A.
(c) The log-rank statistic is $(O_A - E_A)^2/V$, where

$$V = \sum_j \frac{n_{Aj} n_{Bj} d_j (n_j - d_j)}{n_j^2 (n_j - 1)}.$$

```
Compute the statistic and compare it with the $\chi^2$ distribution on one degree
of freedom.
```

(d) Interpret the result.

Solution.

(a) Event times 2, 4, 5, 7, 9, 11, 13 and 17. At risk (n_A, n_B): at 2: $(4, 4)$; at 4: $(4, 3)$; at 5: $(4, 2)$; at 7: $(3, 2)$; at 9: $(3, 1)$; at 11: $(2, 1)$; at 13: $(2, 0)$; at 17: $(1, 0)$.

(b) With one death at each time, $E_A = n_A/(n_A + n_B)$. For group A the observed and expected deaths are:

t	2	4	5	7	9	11	13	17
O_A	0	0	1	0	1	0	1	1
E_A	0.500	0.571	0.667	0.600	0.750	0.667	1.000	1.000

So $O_A = 4$ and $E_A = 5.755$.

(c) The variance contributions are $0.250, 0.245, 0.222, 0.240, 0.188, 0.222$ and 0 at the last two times, giving $V = 1.367$. The statistic is $(4 - 5.755)^2/1.367 = 3.08/1.367 = 2.25$ on 1 degree of freedom, giving $p \approx 0.13$.

(d) Group A died less than expected under the null (4 observed vs 5.76 expected), so group A appears to have somewhat better survival, but the difference is not statistically significant ($\chi^2 = 2.25$, $p \approx 0.13$) — unsurprising with only four patients per group.

3.5 Stratifying a comparison

Type: From research question to model

A multicentre trial compares two chemotherapy treatments for lung cancer. Survival times are available from several centres, and age is also recorded.

(a) Why might you stratify the comparison by centre rather than simply ignore centre?
(b) What does a stratified log-rank test provide that separate (per-centre) tests do not?
(c) In words, what is the null hypothesis of the stratified test?

Solution.

(a) Survival may differ systematically between centres (or age groups); ignoring these variables could confound the treatment comparison. Stratifying accounts for them.
(b) It combines the information across strata into a single overall test of the treatment effect, giving a more precise summary than a set of separate tests.
(c) No treatment difference, after allowing for the stratifying variable (centre or age group).

3.6 A misleading comparison

Type: What's wrong here?

A researcher compares two treatment groups by computing, for each group, the proportion of patients still alive at the end of the study, and then compares these two proportions with a chi-squared test. Patients entered the study at different times and were followed for different lengths of time.

Explain what is wrong with this approach.

Solution. The approach ignores censoring and the actual survival times: patients followed for different durations are treated as if they all had the same follow-up, so the information in the survival times (and in the censored observations) is thrown away. A group followed for a shorter time would misleadingly appear to have worse survival. The comparison should use the full survival times and handle censoring, as the log-rank test does.

4 Mini-lab: Kaplan–Meier and the log-rank test in R

4.1 Comparing two groups in R

Type: Mini-lab

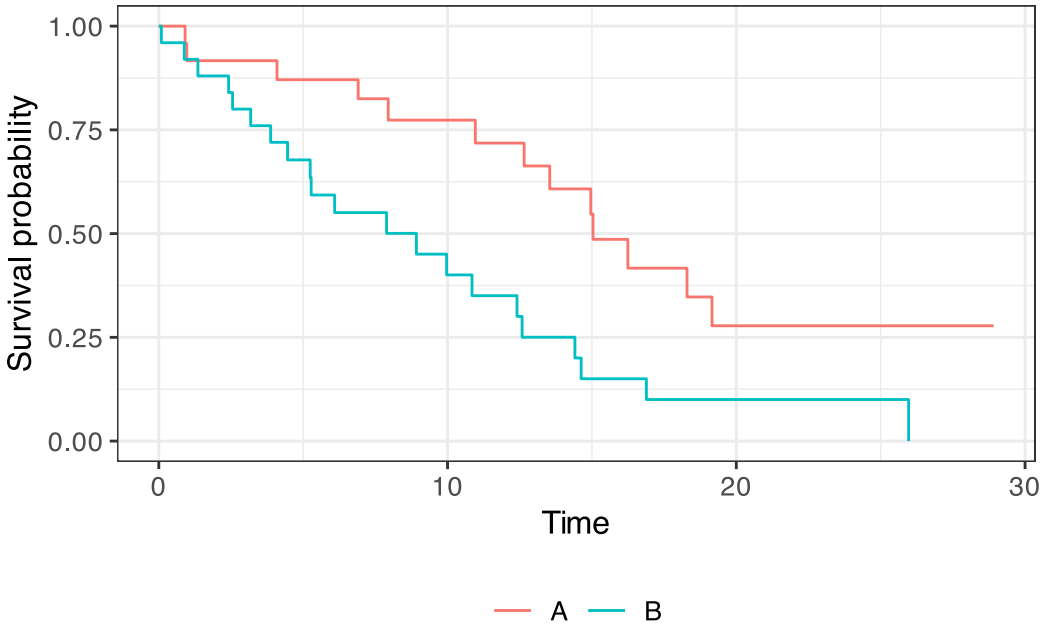
Run the code below. It simulates two groups of 25 patients (Group A with a lower hazard than Group B) with independent censoring, fits Kaplan–Meier curves, and runs a log-rank test.

```
set.seed(7)
n <- 25
time  <- c(rexp(n, 0.05), rexp(n, 0.10))   # Group A then Group B
cens  <- runif(2 * n, 0, 40)
obs   <- pmin(time, cens)
```

```r
event <- as.numeric(time <= cens)
grp   <- factor(rep(c("A", "B"), each = n))
d <- data.frame(time = obs, status = event, group = grp)

library(survival)
library(ggsurvfit)

survfit2(Surv(time, status) ~ group, data = d) |>
  ggsurvfit() +
  labs(x = "Time", y = "Survival probability")
```




```r
survdiff(Surv(time, status) ~ group, data = d)
```

```
Call:
survdiff(formula = Surv(time, status) ~ group, data = d)

         N Observed Expected (O-E)^2/E (O-E)^2/V
group=A 25       13     20.6      2.83      7.49
group=B 25       21     13.4      4.36      7.49

 Chisq= 7.5  on 1 degrees of freedom, p= 0.006
```

(a) From the plot, which group appears to have better survival? Estimate each group's median survival time.
(b) What is the value of the log-rank test statistic, its degrees of freedom and its p-value? State the null hypothesis and your conclusion.

Solution.

(a) Group A lies above Group B, so A has better survival; the printed medians are about 15 for A and about 9 for B.
(b) $\chi^2 = 7.5$ on 1 degree of freedom, $p = 0.006$. The null hypothesis is no difference in survival between the groups. There is strong evidence against it ($p < 0.01$), so we conclude the two groups differ in their survival experience.

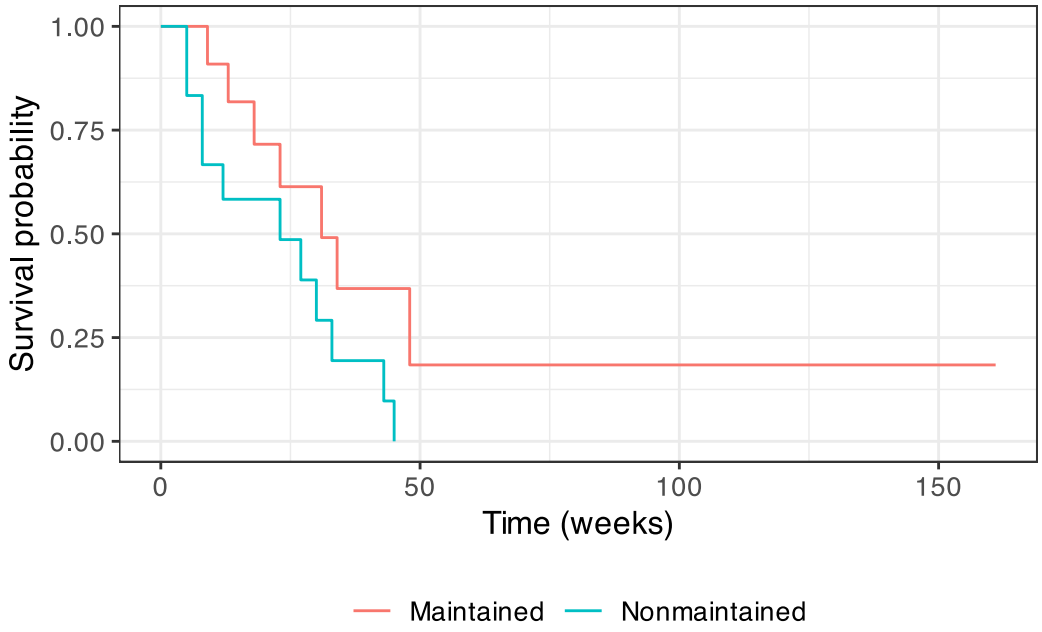
4.2 Analysing a real dataset: acute myeloid leukaemia

Type: Mini-lab

The `aml` dataset (in the `survival` package) contains survival times for 23 patients with acute myeloid leukaemia, grouped by whether they received maintenance chemotherapy (`x`: "Maintained" or "Nonmaintained"). Run the code below to plot the Kaplan–Meier curves and run a log-rank test.

```
library(survival)
library(ggsurvfit)
data(aml)

survfit2(Surv(time, status) ~ x, data = aml) |>
  ggsurvfit() +
  labs(x = "Time (weeks)", y = "Survival probability")
```




```
survdiff(Surv(time, status) ~ x, data = aml)
```

```
Call:
survdiff(formula = Surv(time, status) ~ x, data = aml)

                N Observed Expected (O-E)^2/E (O-E)^2/V
x=Maintained   11        7    10.69      1.27       3.4
x=Nonmaintained 12       11     7.31      1.86       3.4

 Chisq= 3.4  on 1 degrees of freedom, p= 0.07
```

(a) How many patients are in each group, and how many events (deaths) occurred in each?
(b) From the plot, which group appears to have better survival? Report each group's estimated median survival time.
(c) Report the log-rank statistic, its degrees of freedom and p-value. State the null hypothesis and your conclusion.

Solution.

(a) 11 patients were "Maintained" (with 7 deaths) and 12 were "Nonmaintained" (with 11 deaths).
(b) The Maintained group lies above the Nonmaintained group, so it has better survival; the estimated median survival times are about 31 weeks (Maintained) and 23 weeks (Nonmaintained).
(c) $\chi^2 = 3.4$ on 1 degree of freedom, $p = 0.07$. The null hypothesis is no difference in survival between the two groups. The Maintained group died less than expected (7 observed vs 10.7

expected), but the evidence is only modest ($p \approx 0.07$, not significant at the 5% level): there is at most weak evidence that maintenance chemotherapy improves survival in these patients.